

P8107 FINAL EXAM

Fall 2024

Name:

UNI:

For this exam you allowed up to four sheets of paper (front and back). No other reference materials may be used. You have up to 3 hours to complete this exam. Be sure to show all your work. You may attach additional pages to your exam paper if needed.

Honor code:

I have not and will not give or receive aid in this examination nor have I concealed any violation of the University Honor Code.

Sign here: _____

1. (20 points) Random variables $X_1, X_2, \dots, X_n$ are iid $N(\mu, \sigma^2)$ and, independently of the X 's, $Y_1, Y_2, \dots, Y_n$ are iid $N(\mu, 2\sigma^2)$. Assuming that μ is a known constant, the MLE of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{2n} \sum_{i=1}^n (X_i - \mu)^2 + \frac{1}{4n} \sum_{i=1}^n (Y_i - \mu)^2.$$

- Calculate the mean and variance of $\hat{\sigma}^2$.
- Is $\hat{\sigma}^2$ unbiased for σ^2 ? Is it consistent? Justify your answers.

2. (20 points) Random variables $Y_1, Y_2, \dots, Y_n$ are iid with pdf $f(y) = \frac{1+\theta}{\theta} y^{1/\theta}$ for $0 < y < 1$ and $\theta > 0$. Find the MLE of θ . (Note: You don't have to check the second derivative for this problem.)

3. (20 points) Random variables $X_1, X_2, \dots, X_m$ are iid $N(\theta, 1)$ and, independently of the X 's, $Y_1, Y_2, \dots, Y_n$ are iid $N(\theta + 1, 1)$. Find a pivotal quantity based on $\bar{X} + \bar{Y}$ and use it to construct a 95% confidence interval for θ .

4. (20 points) A random variable $Y \sim U(3, 3\theta)$ and it is of interest to test $H_0: \theta = 4$ vs. $H_a: \theta < 4$.
- The test has the form: Reject H_0 if $Y < c$. Find c such that the test has Type I error rate 0.05.
 - Calculate an expression (in terms of θ) for the power of the test in (a). What is the power when $\theta = 3$?

5. (20 points) An estimator of $\mu_1 - \mu_2$ is to be based on $\bar{X} - \bar{Y}$, where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\bar{Y} = \frac{1}{2n} \sum_{i=1}^{2n} Y_i$ (there are two times as many Y 's as X 's). The X 's are independent with mean μ_1 and variance 9, and the Y 's are independent with mean μ_2 and variance 9. (And the X 's are independent of the Y 's.)
- Provide an expression for an approximate 90% confidence interval for $\mu_1 - \mu_2$ based on these data.
 - How large does n need to be for the width of the interval (upper limit minus lower limit) to be no more than 3?